

CGM Glucose Forecasting on OhioT1DM and Shanghai_T1DM: Combined Findings Report

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Datasets: OhioT1DM, all 12 patients (6 from the 2018 release: 559, 563, 570, 575, 588, 591; 6 from the 2020 release: 540, 544, 552, 567, 584, 596) — and Shanghai_T1DM, 12 patients (1001-1012)
Horizons: 15 / 30 / 60 minutes ahead **Lookback window:** 60 minutes (12 steps at 5-minute sampling for Ohio; 4 steps at 15-minute sampling for Shanghai — same wall-clock window, different step count) **Repository:** CGM-Forecasting-Research

This report combines three experiments: (1) a **feature-set ablation** on OhioT1DM (does adding heart rate, meals, sleep, or insulin data to a fixed LSTM improve forecasts?), (2) an **architecture comparison** on OhioT1DM (does a different model — GRU, BiLSTM, TCN, Transformer — outperform a plain LSTM on the best available feature set?), and (3) the same **architecture comparison replicated on the Shanghai_T1DM cohort**, testing whether the best-performing architecture generalizes across datasets, populations, and sampling resolutions. Patient-wise results and mean $\pm$ SD summaries are included throughout, so an average never hides a patient-level outlier without being flagged.

1. Executive Summary

- **Auxiliary signals (HR, carbs, sleep, insulin) give at most a small, mostly non-significant boost** over glucose-only forecasting at 15–30 min horizons on OhioT1DM. The one clear exception: **meal/carb data significantly improves 60-min forecasts** ($p=0.021$).
- **Architecture matters more than extra features did.** Swapping the model architecture (holding a fixed feature set) produced larger, statistically robust differences than swapping feature sets did.
- **GRU is the best-performing and safest architecture at every horizon tested on OhioT1DM**, significantly beating LSTM, BiLSTM, TCN, and Transformer at 15/30 min, and beating BiLSTM/TCN/Transformer (but tying LSTM) at 60 min.
- **GRU's advantage replicates on Shanghai_T1DM — and holds even more consistently.** GRU significantly beats LSTM at all three horizons on Shanghai ($p=0.016, 0.001, 0.043$), where on Ohio the GRU-LSTM gap disappeared at 60 min. This is a genuine cross-dataset generalizability result, not an Ohio-specific artifact.

- **TCN and Transformer underperform simple recurrent models on both datasets** — a clean, consistent, and clinically meaningful finding given the limited lookback window and cohort sizes.
- **One Shanghai patient (1012) shows an architecture-ranking reversal across horizons:** GRU is dramatically the best architecture for this patient at 15/30 min, but becomes the *worst* architecture at 60 min. This is flagged as a genuine anomaly worth investigating, not averaged away.
- **Two OhioT1DM patients (540, 567) are consistently the hardest to forecast** across every architecture and horizon, and patient 540 also has the worst clinical-safety numbers. This should not be averaged away either.
- **Shanghai shows lower RMSE but higher MARD than Ohio at every horizon** — likely driven by Shanghai’s cleaner data (near-zero missingness) combined with shorter per-patient recordings and a single 80/20 split rather than Ohio’s dedicated train/test files, producing more relative-error noise despite better absolute accuracy.

2. Experiment 1 — Feature-Set Ablation on OhioT1DM (fixed LSTM architecture)

Protocol: a single LSTM (64 units) trained separately per patient, per horizon, per feature set. Feature sets B, D, F depend on HeartRate/Steps, which are 100% missing for all 6 2020 patients (different wristband hardware — confirmed by directly inspecting the XML, zero `basis_heart_rate/basis_steps` events) — so those three feature sets ran on the 6 2018 patients only. Feature sets A, C, E ran on all 12.

Feature Set	Channels
A	Glucose only
B	Glucose + HeartRate (<i>2018 only, n=6</i>)
C	Glucose + Carbs (meals)
D	Glucose + HR + Steps + Carbs + Sleep (<i>2018 only, n=6</i>)
E	Glucose + Insulin (bolus + basal)
F	Full multimodal: HR + Steps + Carbs + Sleep + Insulin (<i>2018 only, n=6</i>)

2.1 Statistical significance (Wilcoxon signed-rank, paired by patient)

Across all 12 patients (A vs. C vs. E):

Horizon	Comparison	p-value	Result
15min	A vs C	0.24	not significant
15min	A vs E	0.63	not significant
30min	A vs C	0.11	not significant
30min	A vs E	0.85	not significant
60min	A vs C	0.021	C significantly

Horizon	Comparison	p-value	Result
			better (carbs help at 60min)
60min	A vs E	0.70	not significant

Within the 6 2018 patients only (does HR/Steps/full multimodal beat glucose-only?):

Horizon	A vs B (HeartRate)	A vs D (full wearable)	A vs F (+ insulin)
15min	p=0.44 (B better 4/6)	p=0.16 (D better 5/6)	p=0.84 (F better 3/6)
30min	p=0.69 (B better 4/6)	p=0.56 (D better 4/6)	p=0.44 (F better 4/6)
60min	p=0.69 (B better 4/6)	p=0.16 (D better 5/6)	p=0.44 (F better 2/6)

Interpretation: none of these 6-patient comparisons reach significance — likely underpowered at n=6 rather than a true null result, given D wins in 5/6 patients at two horizons. “D: full wearable + meals + sleep” is the most promising direction worth revisiting with more patients, but should not yet be claimed as proven.

3. Experiment 2 — Architecture Comparison on OhioT1DM (fixed feature set)

Protocol: LSTM, GRU, BiLSTM, TCN, and Transformer, each trained separately per patient, per horizon, on Glucose + carbs + sleep + insulin (the feature subset available for all 12 patients). All 12 patients included in every architecture/horizon cell.

3.1 Summary (mean $\pm$ SD across all 12 patients)

Horizon	Architecture	RMSE	MAE	MARD
15min	GRU	17.32 $\pm$ 2.75	12.00 $\pm$ 1.53	6.99 $\pm$ 1.26
15min	LSTM	18.81 $\pm$ 3.40	13.21 $\pm$ 2.51	8.00 $\pm$ 1.64
15min	BiLSTM	18.71 $\pm$ 3.16	13.12 $\pm$ 2.09	7.35 $\pm$ 1.51
15min	TCN	24.89 $\pm$ 5.85	18.15 $\pm$ 4.52	8.87 $\pm$ 1.29
15min	Transformer	31.04 $\pm$ 4.66	22.85 $\pm$ 3.20	16.16 $\pm$ 3.55
30min	GRU	24.89 $\pm$ 3.35	17.75 $\pm$ 2.18	11.61 $\pm$ 2.28
30min	LSTM	25.96 $\pm$ 3.85	18.80 $\pm$ 2.73	11.92 $\pm$ 2.38
30min	BiLSTM	26.16 $\pm$ 3.90	19.03 $\pm$ 2.71	11.71 $\pm$ 2.49
30min	TCN	31.48 $\pm$ 4.20	23.02 $\pm$ 3.27	13.39 $\pm$ 2.79
30min	Transformer	36.58 $\pm$ 5.17	27.33 $\pm$ 3.80	18.69 $\pm$ 4.24
60min	GRU	36.78 $\pm$ 4.50	27.63 $\pm$ 3.51	17.81 $\pm$ 3.58
60min	LSTM	37.40 $\pm$ 5.04	28.01 $\pm$ 3.83	18.28 $\pm$ 3.85
60min	BiLSTM	38.01 $\pm$ 5.21	28.46 $\pm$ 3.95	19.11 $\pm$ 4.14
60min	TCN	41.18 $\pm$ 4.32	31.24 $\pm$ 3.58	20.04 $\pm$ 4.15

Horizon	Architecture	RMSE	MAE	MARD
60min	Transformer	44.38 ± 6.18	34.03 ± 5.07	24.31 ± 4.88

3.2 Statistical significance (Wilcoxon signed-rank, GRU vs. each other architecture, n=12)

Horizon	GRU vs LSTM	GRU vs BiLSTM	GRU vs TCN	GRU vs Transformer
15min	p=0.0015 ** (11/12)	p=0.0015 ** (11/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)
30min	p=0.0210 * (10/12)	p=0.0015 ** (11/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)
60min	p=0.38 (n.s., 7/12)	p=0.0161 * (10/12)	p=0.0005 *** (12/12)	p=0.0005 *** (12/12)

(* $p<0.05$, ** $p<0.01$, *** $p<0.001$)

Interpretation: GRU's advantage over TCN and Transformer is large, consistent, and highly significant at every horizon. GRU's edge over LSTM and BiLSTM is real but smaller, and **disappears entirely between GRU and LSTM at 60 min** ($p=0.38$) — at longer horizons on Ohio, the two are statistically indistinguishable.

3.3 Clinical safety — Clarke Error Grid, dangerous zones (D + E)

Horizon	GRU	LSTM	BiLSTM	TCN	Transformer
15min	1.11%	1.46%	1.41%	2.44%	2.84%
30min	2.26%	2.46%	2.60%	2.96%	3.75%
60min	4.04%	4.29%	4.05%	4.84%	5.44% (worst)

GRU has the lowest dangerous-zone rate at every horizon (tied with BiLSTM at 60min). Transformer is worst at every horizon, reaching **11.8% dangerous-zone predictions for patient 540 at 60min**.

3.4 Per-patient consistency (OhioT1DM, GRU)

Range at 60 min: 31.1 (patient 596) to 44.4 mg/dL (patient 540) — a 13.3 mg/dL spread.

Consistently hardest patients across every architecture and horizon: 540 and 567 (both 2020 cohort). No systematic 2018-vs-2020 cohort gap was found (Mann-Whitney $p=0.39$ – 0.82 across horizons) — the difficulty is patient-specific, not cohort-specific.

4. Experiment 3 — Cross-Dataset Generalizability: Architecture Comparison on Shanghai_T1DM

Protocol: the same five architectures (LSTM, GRU, BiLSTM, TCN, Transformer), same horizons (15/30/60 min) and same 60-minute lookback window, replicated on the 12 Shanghai_T1DM

patients (1001-1012, each patient's earliest recorded visit). Feature set is Glucose-only, since Shanghai's insulin/diet columns are sparse event markers rather than the dense per-timestep channels available for Ohio — this also makes it directly comparable to Ohio's "A: Glucose only" baseline. Each patient's single longest contiguous CGM segment was split chronologically 80/20 into train/test (Shanghai does not ship separate train/test files the way OhioT1DM does).

4.1 Summary (mean $\pm$ SD across 12 patients, GRU)

Horizon	RMSE	MARD	CEGA_A (accurate zone)
15min	14.59 $\pm$ 4.56	9.44 $\pm$ 3.95%	88.58%
30min	20.51 $\pm$ 4.58	13.98 $\pm$ 5.69%	79.39%
60min	33.08 $\pm$ 12.51	22.51 $\pm$ 9.61%	62.60%

4.2 Statistical significance (Wilcoxon signed-rank, GRU vs. each other architecture, n=12)

Horizon	GRU vs LSTM	GRU vs BiLSTM	GRU vs TCN	GRU vs Transformer
15min	p=0.0161 * (10/12)	p=0.0771 (n.s., 10/12)	p=0.0049 ** (10/12)	p=0.0024 ** (11/12)
30min	p=0.0010 *** (11/12)	p=0.0015 ** (11/12)	p=0.0269 * (9/12)	p=0.0005 *** (12/12)
60min	p=0.0425 * (10/12)	p=0.0771 (n.s., 9/12)	p=0.0034 ** (11/12)	p=0.0005 *** (12/12)

GRU's advantage over LSTM is significant at all three horizons on Shanghai — notably stronger replication than Ohio, where the GRU-LSTM gap vanished at 60 min. GRU vs. TCN/Transformer remains large and significant throughout, matching the Ohio pattern closely.

4.3 Ohio vs. Shanghai, direct comparison (GRU)

Horizon	Ohio RMSE	Shanghai RMSE	Ohio MARD	Shanghai MARD
15min	17.32 $\pm$ 2.75	14.59 $\pm$ 4.56	6.99%	9.44%
30min	24.89 $\pm$ 3.35	20.51 $\pm$ 4.58	11.61%	13.98%
60min	36.78 $\pm$ 4.50	33.08 $\pm$ 12.51	17.81%	22.51%

Shanghai's RMSE is lower than Ohio's at every horizon — plausibly reflecting its much cleaner CGM data (near-zero missingness vs. Ohio's real sensor gaps). But Shanghai's MARD is higher throughout, and its 60-min SD is nearly 3x Ohio's (12.51 vs. 4.50). Likely drivers: Shanghai recordings are much shorter per patient (7–14 days vs. Ohio's multi-week), giving each model far less training data, and the 80/20 chronological split (vs. Ohio's dedicated, separately-collected test files) is a less rigorous held-out evaluation.

4.4 A genuine anomaly: patient 1012's architecture-ranking reversal

Patient 1012 does not follow the overall pattern. At 15/30 min, GRU is *dramatically* the best architecture for this patient (16.21 vs. LSTM's 26.15 at 15min; 25.48 vs. LSTM's 51.00 at 30min). But at 60 min, **GRU becomes the worst-but-one architecture for this same patient** (RMSE 63.56 — only TCN's 67.91 is worse, and it's now behind Transformer's 64.01 and every recurrent alternative). This is a genuine within-patient architecture-ranking reversal across horizons, not a data artifact as far as we can tell from the aggregate pipeline — it warrants direct inspection of this patient's raw CGM trace and predictions before drawing further conclusions from the Shanghai results in isolation. Patient 1005 is also a consistently poor performer across all architectures and horizons (RMSE 48-57 at 60min), though without the same ranking reversal.

5. Combined Interpretation

Putting all three experiments together: **architecture choice mattered more than feature-set choice on Ohio, and the specific winning architecture (GRU) generalizes to a second, independently-collected dataset with a different population, different sampling rate, and different data quality profile.** This is a stronger result than either experiment alone would suggest — it's not just "GRU happened to fit Ohio's 6 2018 patients well," it's a genuine architecture-level finding that replicates.

That said, the Shanghai replication comes with real caveats the report should not gloss over: higher relative error (MARD) despite lower absolute error (RMSE), much larger patient-to-patient variance at 60 min, and one clear anomalous patient (1012) whose architecture ranking flips entirely at the longest horizon. Before treating "GRU generalizes" as a settled conclusion, it's worth (a) investigating patient 1012 directly, and (b) re-running the Shanghai comparison with a more rigorous train/test split (e.g. a dedicated held-out period per patient, rather than a chronological 80/20 split) to rule out that Shanghai's higher variance is a data-splitting artifact rather than a genuine population difference.

6. Known Limitations

- **Feature-set comparisons for B/D/F run on only 6 patients** (2018 cohort) — likely underpowered; the promising "D: full wearable" direction needs a larger sample before being treated as established.
- **CEGA zone classification and MARD/time-lag are only computed in Experiments 2 and 3** (architecture comparisons) — Experiment 1 (feature-set) only reports RMSE/MAE/R². A useful follow-up would be re-running the feature-set ablation with the full clinical metric suite.
- **N-BEATS/N-HiTS and TFT were not included** in either architecture comparison — N-BEATS needs a univariate-only variant, and TFT needs a different stack (pytorch-forecasting) than the Keras models used here.
- **Shanghai uses a single 80/20 chronological split per patient**, not a dedicated separately-collected test set like Ohio — a genuine methodological gap between the two experiments that likely contributes to Shanghai's higher variance (see §4.3-4.4).

- **Shanghai feature set is glucose-only** — an insulin-inclusive comparison (mirroring Ohio's bolus/basal extraction) has not yet been built for Shanghai.
 - **Single train/val/test split per patient in all experiments** — no cross-validation; results reflect one held-out test period per patient, not a K-fold estimate.
 - **Patient 1012's anomalous 60-min result has not been root-caused** — flagged in §4.4, not yet investigated further.
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Appendix: Data Files

- `featureset_patient_wise.csv` — full per-patient results, Experiment 1 (OhioT1DM feature-set ablation)
- `architecture_patient_wise.csv` — full per-patient results, Experiment 2 (OhioT1DM architecture comparison)
- `shanghai_t1dm_architecture_results.csv` — full per-patient results, Experiment 3 (Shanghai_T1DM architecture comparison)
- `feat_summary.csv` / `arch_summary.csv` / `shanghai_t1dm_architecture_summary.csv` — mean $\pm$ SD summary tables in machine-readable form
- `gru_patient_wise.csv` — the per-patient OhioT1DM GRU breakdown used in §3.4
- `architecture_comparison_chart.png` — mean RMSE by architecture across horizons (OhioT1DM)
- `per_patient_rmse.png` — per-patient RMSE spread, glucose-only baseline (OhioT1DM)